Predicting the next-day direction of WEGE3.SA

MSDS 451, Programming Assignment 1. The asset is WEG ON, ticker WEGE3.SA.

1. Problem

The task is to predict the direction of the next session's close-to-close log return. The series is WEGE3.SA, the common share of WEG S.A. listed on B3. It is not the WTI series used in the course example.

On day t the label is 1 when log(C(t) / C(t-1)) is positive, and 0 when the close is flat or down. A forecast for day t may use only prices and volume known at the close of day t-1. The open, high, low, close, and volume of day t itself are not predictors.

The course example fits an XGBoost classifier and then scores that fit on the same rows used to train it. This report keeps a later block of sessions out of feature selection and out of the hyperparameter search. The finished classifier is scored on that block once.

2. Data and research design

Daily open, high, low, close, and volume come from Yahoo Finance for WEGE3.SA. Yahoo applies the split and dividend adjustment. The request covers 2000-01-01 through 2026-10-11. Yahoo treats the end date as exclusive, so the last session in the file is 2026-10-09. The download has 6,721 sessions.

The program is the numbered scripts in pipeline/. Shared settings are in 01_config.py. The later scripts download prices, build 25 candidate columns, cut the sample, select a subset by AIC, tune XGBoost, score the hold-out once, and render this report. Python cannot import a module whose name starts with a digit, so 09_run.py loads each file by name.

A row enters the feature table only when all 25 candidates are defined. That table runs from 2007-07-03 through 2026-10-09 and has 4,418 sessions. It is then cut with two settings in 01_config.py.

DEV_END is 2023-12-31. Development is every feature row on or before that date: 3,723 sessions, from 2007-07-03 through 2023-12-28. The next 10 sessions, 2024-01-02 through 2024-01-15, are a gap. They are neither fit nor scored. The remaining 685 sessions, 2024-01-16 through 2026-10-09, are the hold-out.

Feature selection and the hyperparameter search read the development file only. After both choices are fixed, XGBoost is fit on all 3,723 development sessions and applied once to the hold-out. The hold-out score does not change the columns or the settings.

3. Features

The target is the direction of the close-to-close log return on WEGE3.SA. On day t the label is 1 when log(C(t) / C(t-1)) is positive, and 0 when the close is flat or down. Each of the 25 candidates is known at the close of day t-1. The open, high, low, close, and volume of day t do not enter any candidate. Values are stored at full precision.

The 25 columns are one pool. A later chapter searches their subsets jointly. The first 15 follow the course example and are measured in reais or in shares. The other 10 restate return, volatility, and liquidity in units that do not grow with the price level. The adjusted close of WEG rises from about 0.09 reais in 2000 to about 55 reais in 2026, so a lag of the price mostly tracks that rise.

Definitions

Feature	Definition, using information through day t-1	Role
CloseLag1	Close on t-1	Example. Yesterday's price level.
CloseLag2	Close on t-2	Example. Price level two sessions back.
CloseLag3	Close on t-3	Example. Price level three sessions back.
HMLLag1	High minus low on t-1	Example. Yesterday's range, in reais.
HMLLag2	High minus low on t-2	Example.
HMLLag3	High minus low on t-3	Example.
OMCLag1	Open minus close on t-1	Example. Yesterday's open-to-close change, in reais.
OMCLag2	Open minus close on t-2	Example.
OMCLag3	Open minus close on t-3	Example.
VolumeLag1	Share volume on t-1	Example. Undefined when the recorded volume is zero.
VolumeLag2	Share volume on t-2	Example. Same zero-volume rule.
VolumeLag3	Share volume on t-3	Example. Same zero-volume rule.
CloseEMA2	Exponential moving average of the close through t-1, half-life of 1 session	Example. The jump-start names are EMA2, EMA4, and EMA8; the half-lives are 1, 2, and 4 sessions.
CloseEMA4	Same average, half-life of 2 sessions	Example.
CloseEMA8	Same average, half-life of 4 sessions	Example.
Ret1	log(C(t-1) / C(t-2))	One-day reversal (Jegadeesh, 1990; Lehmann, 1990).
Ret21	log(C(t-1) / C(t-22))	About one month of past return.
Mom12_1	log(C(t-21) / C(t-252))	Twelve-month return, skipping the most recent month (Jegadeesh and Titman, 1993).
RVol21	Sample standard deviation of the 21 daily log returns ending on t-1	Recent volatility.
Parkinson21	Mean, over those 21 sessions, of [log(H/L)]^2 / (4 ln 2)	Range-based variance (Parkinson, 1980). It uses the high and the low.
LogVolChg1	log(V(t-1) / V(t-2)), on positive volume	Volume shock, linked to short-horizon reversal (Campbell, Grossman, and Wang, 1993).
Amihud21	Mean, over the 21 sessions ending on t-1, of absolute return / (close times share volume)	Illiquidity (Amihud, 2002). Each session in the window must have positive volume.
Range1	(H(t-1) - L(t-1)) / C(t-1)	Yesterday's range as a fraction of price. Scale-free HML.
Intraday1	log(C(t-1) / O(t-1))	Yesterday's open-to-close return. Scale-free OMC.
Overnight1	log(O(t-1) / C(t-2))	Yesterday's overnight gap.

Volume and the estimation sample

Yahoo records volume of zero on 1,814 of the 6,721 downloaded sessions, including nearly every session from 2000 through 2006. A zero is treated as missing volume, not as a session with no shares traded. Volume lags, the log volume change, and the Amihud ratio are undefined on a zero print. A row is kept only when all 25 candidates are defined. Closes from before the first kept row still enter the moving averages and Mom12_1. They are history, not rows of their own. No missing value is filled in.

The kept table has 4,418 sessions, from 3 July 2007 through 9 October 2026. Of these, 2,229 (50.5 percent) are up days and 2,189 are flat or down. Within that span, 368 downloaded sessions are absent because a zero-volume print leaves at least one candidate undefined.

The same file stores the day's raw prices, the same-day range, the same-day open-minus-close, and the log return, so the candidates can be checked. Those columns are not part of the 25.

RSI, MACD, Bollinger bands, fractional differentiation, and any series that is not in this file of daily open, high, low, close, and volume are outside the candidate list. Chapter 2 cuts this table into a development window and a hold-out window.

4. Feature selection

The search uses the 3,723 development sessions only. Each candidate subset is fit by an unpenalized logistic regression, which is the maximum likelihood the AIC formula assumes. The subset with the smallest AIC is kept. Columns are standardized with the mean and standard deviation of those development sessions. For an unpenalized logistic regression with an intercept, that rescaling does not change the maximized likelihood. The hold-out file is not read.

Combinations that repeat the same price

Six columns track one price path: CloseLag1, CloseLag2, CloseLag3, CloseEMA2, CloseEMA4, and CloseEMA8. On the development window the smallest absolute correlation among them is 0.999. Adding the other five to CloseLag1 raises the log-likelihood by 1.45. AIC needs a gain above 1 for each added column, so a second copy does not pay for itself. CloseLag1 with CloseLag3 gains 0.26. CloseLag1 with CloseEMA8 gains 0.00.

The search allows at most one of those six. The other 19 columns stay free, including next to that one price column. High minus low together with the close still belongs in the search: that pair gains 1.15 over high minus low alone, which clears the hurdle of 1.

Subsets of all 25 columns number 33,554,431. With at most one price-level column the number is 3,670,015. A second price-level column cannot improve the AIC, so the cut does not change the winner.

Selected subset

The smallest AIC is 5157.46, on VolumeLag1, Ret1, CloseEMA8. An intercept-only model on the same 3,723 sessions has AIC 5162.88. The winner improves on that null model by 5.42.

Rank	AIC	Features
1	5157.46	VolumeLag1, Ret1, CloseEMA8
2	5157.58	VolumeLag1, Ret1, CloseEMA4
3	5157.61	VolumeLag1, Ret1, CloseLag3
4	5157.64	VolumeLag1, Ret1, CloseEMA2
5	5157.66	VolumeLag1, Ret1, CloseLag2
6	5157.66	VolumeLag1, Ret1, CloseLag1
7	5158.00	VolumeLag1, VolumeLag3, Ret1, CloseEMA8
8	5158.13	VolumeLag1, VolumeLag3, Ret1, CloseEMA4
9	5158.17	VolumeLag1, VolumeLag3, Ret1, CloseLag3
10	5158.19	VolumeLag1, VolumeLag3, Ret1, CloseEMA2

The search scored 3,670,015 subsets in 35.3 minutes. 47,799 fits did not settle and were discarded.

5. XGBoost

The classifier uses the three columns chosen in the previous chapter: VolumeLag1, Ret1, CloseEMA8. Optuna runs 100 trials on the 3,723 development sessions. Each trial is a different combination of tree depth, minimum child weight, row subsample, learning rate, and number of trees. The next trial is aimed near combinations that

already scored well. The score is the mean ROC AUC across 5 time-series folds, with a gap of 10 sessions. The subsample is kept between 0.5 and 1. The seed is 21. The hold-out file is not read.

The best mean AUC on the development folds is 0.5335, with a spread across folds of 0.0307. The chosen settings are depth 9, minimum child weight 9, subsample 0.595, learning rate 0.013, and 925 trees.

Ten best trials

Rank	Mean AUC	Depth	Min child	Subsample	Learning rate	Trees
1	0.5335	9	9	0.595	0.013	925
2	0.5334	8	9	0.576	0.017	881
3	0.5323	9	9	0.573	0.034	676
4	0.5321	8	9	0.555	0.023	845
5	0.5318	9	9	0.513	0.018	889
6	0.5318	9	9	0.592	0.012	991
7	0.5317	9	9	0.548	0.034	753
8	0.5315	9	9	0.573	0.014	996
9	0.5312	8	8	0.524	0.016	847
10	0.5310	9	8	0.555	0.025	791

6. Hold-out

The classifier is fit once on the 3,723 development sessions, from 2007-07-03 through 2023-12-28. The settings are the ones chosen in the previous chapter: depth 9, minimum child weight 9, subsample 0.595, learning rate 0.013, and 925 trees. The columns are VolumeLag1, Ret1, CloseEMA8. The seed is 21. The fit does not see the hold-out.

The score uses the 685 hold-out sessions, from 2024-01-16 through 2026-10-09. The 10 sessions from 2024-01-02 through 2024-01-15 stay out of the fit and out of the score. A session is called up when the fitted probability is at least 0.5. The development cross-validation AUC was 0.5335. That number is not recomputed here.

Two reference rules use no fitted parameters. Always up predicts an up day every session. Repeat previous direction predicts that today's direction equals the direction of the immediately previous session. For the first hold-out day, that previous session is 2024-01-15.

Down hit counts down-or-flat days predicted as down-or-flat. Down miss counts those days predicted as up. Up miss counts up days predicted as down-or-flat. Up hit counts up days predicted as up. The always-up rule assigns the same score to every session, so its ROC AUC is 0.5.

Hold-out scores

Rule	Accuracy	ROC AUC	Down hit	Down miss	Up miss	Up hit
XGBoost	0.4847	0.4752	261	76	277	71
Always up	0.5080	0.5000	0	337	0	348
Repeat previous direction	0.4599	0.4597	152	185	185	163

The hold-out share of up days is 0.5080. Always up is therefore right on 0.5080 of the hold-out sessions. The fitted classifier is right on 0.4847, with ROC AUC 0.4752. Repeat previous direction is right on 0.4599, with ROC AUC 0.4597.

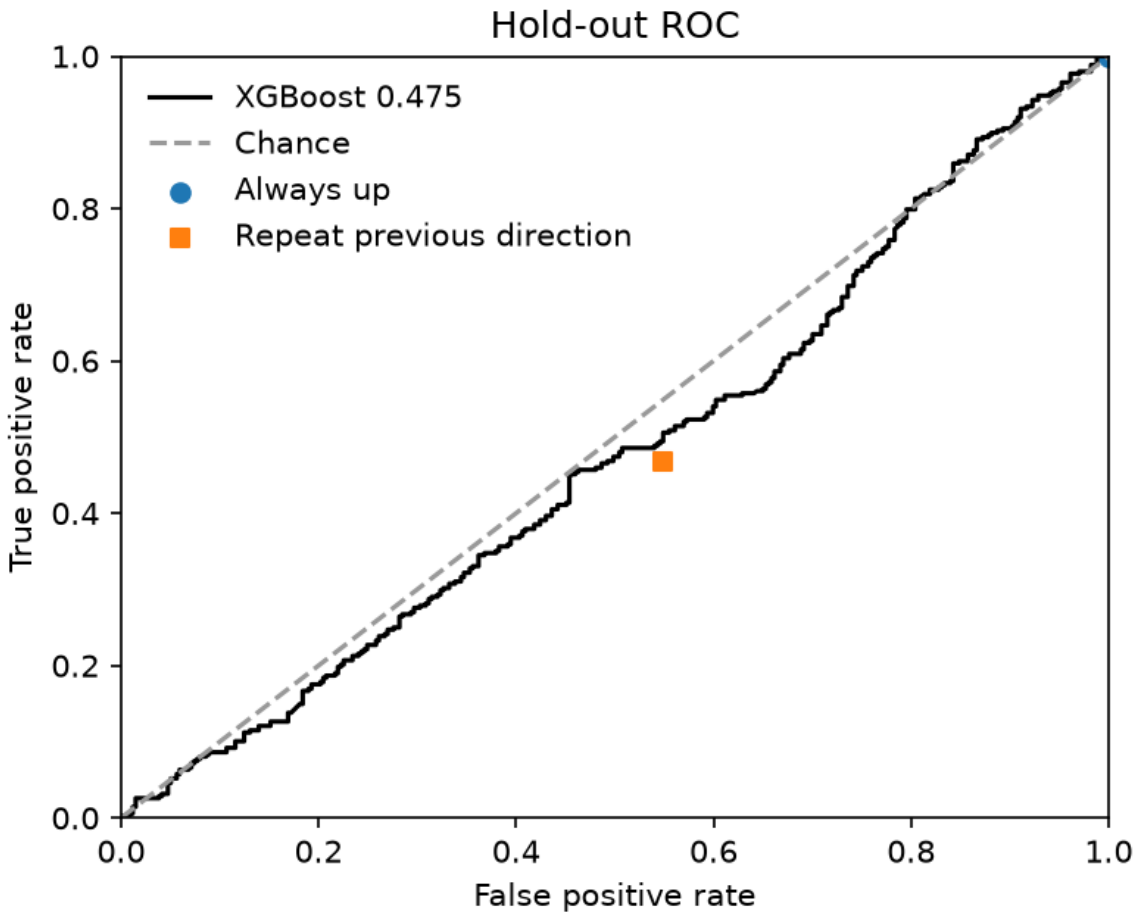


Figure 1. Hold-out ROC curve. The dashed line is chance.

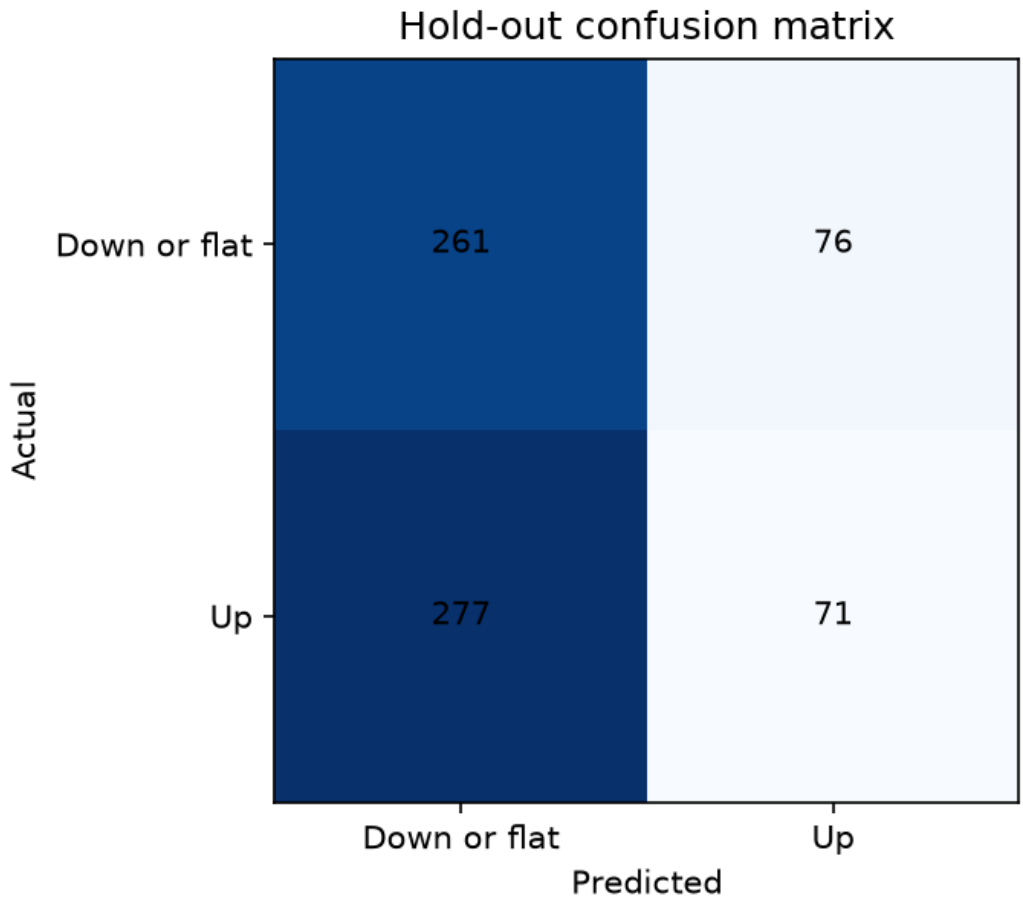


Figure 2. Hold-out confusion matrix for the fitted classifier.

7. Conclusion

The selected logistic model is a small improvement on an intercept. Its AIC is 5157.46. An intercept-only model on the same development sessions has AIC 5162.88. The XGBoost search, scored by mean ROC AUC on five time-series folds of the development window, reaches 0.5335.

On the 685 hold-out sessions the fitted classifier has accuracy 0.4847 and ROC AUC 0.4752. Always predicting an up day has accuracy 0.5080. Repeating the previous session's direction has accuracy 0.4599. On sessions the model did not see, the constant up forecast is ahead of the fitted classifier.

The course example reports classification accuracy of 81.8 percent and an AUC of 0.82 after fitting and scoring the same rows. Its time-series cross-validation accuracy was about 50 to 51 percent, and the example states that training-set predictions are better than should be expected on new data. The hold-out numbers in this report sit with that cross-validation result.

References

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Prices were downloaded with yfinance. Candidate columns were built with Polars. The AIC search checks its fit against scikit-learn's unpenalized logistic regression. The classifier is XGBoost, and the hyperparameter search is Optuna. The figures use Matplotlib.